Global convergence of the inner iterations with the exact projected gradient path

Traulls.jl — working notes for the COAP revision

October 24, 2026

Abstract

These notes prove the global convergence of the inner trust-region iterations (Algorithm 2 of the manuscript) when the Cauchy step follows the exact projected gradient path over the polyhedron $\Omega \cap \{\|s\|_\infty \le \Delta\}$. They show that the first local minimizer of the model along this path has no uniform decrease guarantee over a polyhedron, identify the bound releases as the cause, and add a one-parameter safeguard under which the standard Cauchy decrease holds with no rank assumption. They complement `cauchy-projected-gradient-path.md`, whose open points 1 and 2 (§9) they settle. The setting is idealized: exact arithmetic, exact path, no tolerances. The parking, tie and noise tolerances of the implementation are not discussed.

Summary

1. **The first local minimizer alone has no uniform decrease guarantee over a polyhedron.** The model decrease it provides cannot be bounded below by any positive function of $h(x) = \|P_\Omega(x - g) - x\|$, $\pi_\varepsilon(x)$, Δ and $\|H\|$. This is not a weakness of the proof. A four-variable example (Section 3.2) has $\Delta = 1$, $H \succeq 0$ with $\|H\| = 2$, $h \to \sqrt{2}/10$ and $\pi_\varepsilon = 0.2$, while the decrease at the first local minimizer is $3.7\delta + O(\delta^2) \to 0$. It is the zigzag example of the manuscript plus one free variable, which turns the stationary piece into a *slow* piece.

2. **The cause is the releases.** Along a piece, the speed $\|s'(t)\|$ is the norm of the reduced gradient. It decreases at a hit and increases at a release (Lemma 2). If no bound is released after the first local minimizer, as on a box, the former argument is correct and gives the decrease of Section 3.1 with no safeguard (Proposition 4). The old draft's error (nested tangent spaces) is exactly the missing release case.

3. **A one-parameter safeguard restores the guarantee.** Let $L(t) = -\langle g, s(t)\rangle$ and let t_G be the first $t > 0$ at which $\psi(t) = q(s(t))$ reaches $-\kappa_G L(t)$, with $\kappa_G \in (0, 1)$. The first local minimizer is kept whenever t_G (or the end of the path) comes first. Otherwise the scan goes on to the next local minimizer after t_G and returns the best one met (Section 4.1). In random tests the step changes by less than 0.7%, and the safeguard costs one extra piece in the median case (Section 7.3).

4. **Cauchy decrease** (Theorem 7), with no rank assumption, no threshold on ε, and no dimension constant:

$$-q(s^C) \;\ge\; \frac{\kappa_G}{2}\, h(x)\, \min\Big(\Delta,\; \frac{\kappa'\, h(x)}{1 + \|H\|}\Big), \qquad \kappa' = \min(1, 4(1 - \kappa_G)).$$

5. **Convergence.** $\liminf_k h(x_k) = 0$ with the proof of `theo:inner_convergence` unchanged. Since h is continuous, the growth assumption on b_k also gives $\lim_k h(x_k) = 0$ (Theorem 9).

6. **Stopping test.** If $h(x) \le \varepsilon$, then $\pi_\varepsilon(x) \le \max(1, 1/\sigma^+)\, h(x)$. Here σ^+ is the smallest nonzero singular value of the matrices $[A; E_S^\top]$, and the multipliers are those of minimal sign violation, which is what the code computes. Hence the test $\pi_\varepsilon(x_k) \le \omega$ is met after finitely many iterations, for every $\varepsilon > 0$ (Corollary 12). The outer analysis does not change.

1 Setting

1.1 Step set, path, model

Fix an inner iterate $x \in \Omega$, its gradient g, a symmetric H and $\Delta > 0$. The steps lie in the polyhedron

$$D = \{s \in \mathbb{R}^n \mid As = 0,\ \ell^\Delta \le s \le u^\Delta\},$$
$$\ell_i^\Delta = \max(-\Delta,\, l_i - x_i) \le 0, \qquad u_i^\Delta = \min(\Delta,\, u_i - x_i) \ge 0,$$

so $0 \in D$, D is bounded, and $D = (\Omega - x) \cap \{\|s\|_\infty \le \Delta\}$. The projected gradient path is $s(t) = P_D(-tg)$, $t \ge 0$, and

$$\psi(t) = q(s(t)) = \langle g, s(t)\rangle + \tfrac{1}{2}\langle s(t), Hs(t)\rangle, \qquad L(t) = -\langle g, s(t)\rangle.$$

L is the decrease of the *linear* model along the path, and $\psi(t) = -L(t) + \frac{1}{2}\langle s(t), Hs(t)\rangle$. The criticality measure is the projected gradient

$$p = P_\Omega(x - g) - x, \qquad h(x) = \|p\|.$$

It is continuous in x, and $h(x) = 0$ if and only if x is first-order critical. We also mention the measure $\chi(x) = -\min\{\langle g, d\rangle \mid x + d \in \Omega,\ \|d\|_\infty \le 1\}$ of [2], which is not needed in the proofs.

1.2 Faces, normal cones and projections

This subsection gathers the facts on polyhedra that the description of the path uses. Since $\Delta < \infty$, all the bounds of D are finite. We write them as pairs: $\beta = (i, -)$ stands for $s_i \ge \ell_i^\Delta$ and $\beta = (i, +)$ for $s_i \le u_i^\Delta$. With

$$a_{(i,-)} = -e_i, \quad b_{(i,-)} = -\ell_i^\Delta, \qquad a_{(i,+)} = e_i, \quad b_{(i,+)} = u_i^\Delta,$$

the bound β reads $\langle a_\beta, s\rangle \le b_\beta$, and $D = \{s \mid As = 0,\ \langle a_\beta, s\rangle \le b_\beta \text{ for all } \beta\}$. For $s \in D$, $\mathcal{A}(s) = \{\beta \mid \langle a_\beta, s\rangle = b_\beta\}$ is the set of bounds active at s. For a set $\mathcal{W}$ of bounds, $I(\mathcal{W}) = \{i \mid (i,-) \in \mathcal{W} \text{ or } (i,+) \in \mathcal{W}\}$ is the set of the variables concerned, and E_I denotes the matrix of the columns e_i, $i \in I$.

Definition.

(a) A *face* of D is a nonempty set of the form

$$F_\mathcal{W} = \{s \in D \mid \langle a_\beta, s\rangle = b_\beta \text{ for all } \beta \in \mathcal{W}\},$$

where $\mathcal{W}$ is a set of bounds. For a face F, we denote by $\mathcal{W}(F) = \{\beta \mid \langle a_\beta, s\rangle = b_\beta \text{ for all } s \in F\}$ the set of the bounds active on all of F, and $I(F) = I(\mathcal{W}(F))$.

(b) The *affine hull* of a nonempty set $S \subseteq \mathbb{R}^n$ is the smallest affine subspace of $\mathbb{R}^n$ containing S, that is, the set of its affine combinations

$$\operatorname{aff} S = \Big\{ \sum_{k=1}^p \lambda_k s_k \;\Big|\; p \ge 1,\ s_k \in S,\ \lambda_k \in \mathbb{R},\ \sum_{k=1}^p \lambda_k = 1 \Big\}.$$

An affine subspace E is the translate $E = c + T$ of a unique linear subspace $T = E - E$, its *direction space*, by any $c \in E$.

(c) The *relative interior* of a convex set S is its interior relative to $\operatorname{aff} S$:

$$\operatorname{ri} S = \{s \in S \mid \exists r > 0,\ \{v \in \operatorname{aff} S \mid \|v - s\| < r\} \subseteq S\}.$$

(d) The *normal cone* to D at $s \in D$ is $N_D(s) = \{w \in \mathbb{R}^n \mid \langle w, v - s\rangle \le 0 \text{ for all } v \in D\}$. As an intersection of closed half-spaces, it is a closed convex cone.

A face F is convex, as the intersection of D with hyperplanes, and $F = F_{\mathcal{W}(F)}$. Indeed, $F = F_{\mathcal{W}}$ for some $\mathcal{W} \subseteq \mathcal{W}(F)$, so that $F_{\mathcal{W}(F)} \subseteq F_{\mathcal{W}} = F$, and $F \subseteq F_{\mathcal{W}(F)}$ by definition of $\mathcal{W}(F)$.

Lemma A (faces)**.** *Let F be a face of D, $\mathcal{W} = \mathcal{W}(F)$ and $I = I(F)$.*

(i) The affine hull of F is

$$\operatorname{aff} F = \{s \in \mathbb{R}^n \mid As = 0,\ \langle a_\beta, s\rangle = b_\beta \text{ for all } \beta \in \mathcal{W}\},$$

i.e. the points with $As = 0$, $s_i = \ell_i^\Delta$ for $(i,-) \in \mathcal{W}$ and $s_i = u_i^\Delta$ for $(i,+) \in \mathcal{W}$. Its direction space is $T_F = \{v \in \mathbb{R}^n \mid Av = 0,\ v_i = 0 \text{ for all } i \in I\}$.

(ii) $\operatorname{ri} F = \{s \in D \mid \mathcal{A}(s) = \mathcal{W}\} = \{s \in \operatorname{aff} F \mid \langle a_\beta, s\rangle < b_\beta \text{ for all } \beta \notin \mathcal{W}\}$. *In particular,* $\operatorname{ri} F$ *is convex and nonempty.*

(iii) Every $s \in D$ belongs to $\operatorname{ri} F$ *for exactly one face F, namely $F = F_{\mathcal{A}(s)}$, and then $\mathcal{W}(F) = \mathcal{A}(s)$.*

(iv) D has finitely many faces.

Proof. We first build a point $\hat{s} \in F$ with $\mathcal{A}(\hat{s}) = \mathcal{W}$. For each bound $\beta \notin \mathcal{W}$, there is $s^\beta \in F$ with $\langle a_\beta, s^\beta\rangle < b_\beta$, by definition of $\mathcal{W}$. Let $\hat{s}$ be the average of these points (any point of F if every bound is in $\mathcal{W}$). Then $\hat{s} \in F$ by convexity. For $\beta \notin \mathcal{W}$, $\langle a_\beta, \hat{s}\rangle$ is the average of numbers $\le b_\beta$, one of which is $< b_\beta$, so $\langle a_\beta, \hat{s}\rangle < b_\beta$. The bounds of $\mathcal{W}$ are active at $\hat{s}$ since $\hat{s} \in F$.

(i) Let E be the right-hand side. It is an affine subspace containing F, so $\operatorname{aff} F \subseteq E$. Conversely, let $e \in E$ and $s_\lambda = \hat{s} + \lambda(e - \hat{s})$ for $\lambda > 0$. Then $s_\lambda \in E$, so $As_\lambda = 0$ and the bounds of $\mathcal{W}$ are active at s_λ. The finitely many strict inequalities $\langle a_\beta, \hat{s}\rangle < b_\beta$, $\beta \notin \mathcal{W}$, still hold at s_λ for λ small enough. Hence $s_\lambda \in D$ and $s_\lambda \in F$. Then $e = (1 - 1/\lambda)\,\hat{s} + (1/\lambda)\,s_\lambda$ is an affine combination of points of F, and $e \in \operatorname{aff} F$. The direction space is $E - E = \{v \mid Av = 0,\ \langle a_\beta, v\rangle = 0\ (\beta \in \mathcal{W})\}$, and $\langle a_\beta, v\rangle = \pm v_i$ for $\beta = (i, \pm)$, which gives T_F.

(ii) Let $s \in D$ with $\mathcal{A}(s) = \mathcal{W}$. Then $s \in F_{\mathcal{W}} = F$. For $v \in \operatorname{aff} F = E$ close to s, the strict inequalities $\langle a_\beta, v\rangle < b_\beta$, $\beta \notin \mathcal{W}$, still hold, and the other constraints of D hold with equality on E. So $v \in F$, and $s \in \operatorname{ri} F$. Conversely, let $s \in F$ and suppose that a bound $\beta \notin \mathcal{W}$ is active at s. For $\lambda > 0$, the point $s + \lambda(s - \hat{s})$ belongs to $\operatorname{aff} F$, tends to s as $\lambda \to 0$, and violates β, since $\langle a_\beta, s + \lambda(s - \hat{s})\rangle = b_\beta + \lambda(b_\beta - \langle a_\beta, \hat{s}\rangle) > b_\beta$. So $s \notin \operatorname{ri} F$. This proves the first equality. For the second, a point of $\operatorname{aff} F$ satisfies $As = 0$ and the bounds of $\mathcal{W}$ with equality. It thus belongs to D with $\mathcal{A}(s) = \mathcal{W}$ if and only if the other bounds hold strictly. This set is the intersection of an affine subspace with open half-spaces, hence convex, and it contains $\hat{s}$.

(iii) Let $s \in D$ and $F = F_{\mathcal{A}(s)}$, a face since $s \in F$. Every $s' \in F$ satisfies the bounds of $\mathcal{A}(s)$ with equality, so $\mathcal{A}(s) \subseteq \mathcal{W}(F)$. Conversely, the bounds of $\mathcal{W}(F)$ are active at $s \in F$, so $\mathcal{W}(F) \subseteq \mathcal{A}(s)$. Hence $\mathcal{W}(F) = \mathcal{A}(s)$, and $s \in \operatorname{ri} F$ by (ii). If $s \in \operatorname{ri} F'$ for another face F', then $\mathcal{W}(F') = \mathcal{A}(s)$ by (ii), and $F' = F_{\mathcal{W}(F')} = F_{\mathcal{A}(s)} = F$.

(iv) There are finitely many sets of bounds. □

Lemma B (normal cones)**.** *Let $s \in D$.*

(i) $N_D(s) = \{A^\top \xi + \sum_{\beta \in \mathcal{A}(s)} \lambda_\beta a_\beta \mid \xi \in \mathbb{R}^m,\ \lambda_\beta \ge 0\}$.

(ii) *Equivalently, $w \in N_D(s)$ if and only if $-w = A^\top \xi + \sum_{i \in I(\mathcal{A}(s))} \eta_i e_i$ for some $\xi \in \mathbb{R}^m$ and η with $\sigma_i \eta_i \geq 0$, where $\sigma_i = +1$ if only the lower bound of i is active, $\sigma_i = -1$ if only the upper bound is, and $\sigma_i = 0$ (no condition) if both are, i.e. $\ell_i^\Delta = u_i^\Delta$.*

(iii) *For a face F, $N_D(s)$ is the same cone at every $s \in \operatorname{ri} F$. We denote it by $N_D(F)$. It is contained in $T_F^\perp = \operatorname{range} \begin{bmatrix} A^\top & E_{I(F)} \end{bmatrix}$.*

Proof. (i) If $w = A^\top \xi + \sum_{\beta \in \mathcal{A}(s)} \lambda_\beta a_\beta$ with $\lambda \geq 0$, then for $v \in D$, $\langle w, v - s \rangle = \langle \xi, A(v - s) \rangle + \sum_{\beta \in \mathcal{A}(s)} \lambda_\beta (\langle a_\beta, v \rangle - b_\beta) \leq 0$, since $Av = As = 0$ and $\langle a_\beta, s \rangle = b_\beta$ for $\beta \in \mathcal{A}(s)$. Conversely, let $w \in N_D(s)$. Then s solves the linear program $\max\{\langle w, v \rangle \mid Av = 0,\ \langle a_\beta, v \rangle \leq b_\beta \text{ for all } \beta\}$. By the duality theorem of linear programming, its dual $\min\{\sum_\beta b_\beta \lambda_\beta \mid A^\top \xi + \sum_\beta \lambda_\beta a_\beta = w,\ \lambda \geq 0\}$ has a solution (ξ, λ) with the same value $\langle w, s \rangle$. Subtracting, $0 = \sum_\beta b_\beta \lambda_\beta - \langle w, s \rangle = \sum_\beta \lambda_\beta (b_\beta - \langle a_\beta, s \rangle)$, a sum of nonnegative terms. So $\lambda_\beta = 0$ for every $\beta \notin \mathcal{A}(s)$.

(ii) Group the two bounds of each variable: $\sum_{\beta \in \mathcal{A}(s)} \lambda_\beta a_\beta = \sum_i (\lambda_{(i,+)} - \lambda_{(i,-)})\, e_i$, with $\lambda_{(i,\pm)} = 0$ when $(i, \pm)$ is inactive. Set $\eta_i = \lambda_{(i,-)} - \lambda_{(i,+)}$ and change ξ into $-\xi$. Then $\eta_i \geq 0$ if only the lower bound is active, $\eta_i \leq 0$ if only the upper bound is, and η_i is arbitrary if both are. Conversely, every such η is obtained in this way.

(iii) By (i), $N_D(s)$ only depends on $\mathcal{A}(s)$, which equals $\mathcal{W}(F)$ on $\operatorname{ri} F$ by Lemma A(ii). By Lemma A(i), $T_F = \{v \mid Av = 0,\ E_{I(F)}^\top v = 0\}$ is the null space of the matrix whose transpose is $[A^\top\ E_{I(F)}]$, so $T_F^\perp$ is the range of $[A^\top\ E_{I(F)}]$. By (ii), every element of $N_D(F)$ lies in this range. □

Lemma C (projections)**.** *Let $y \in \mathbb{R}^n$ and let F be a face of D.*

(i) *$s = P_D(y)$ if and only if $s \in D$ and $y - s \in N_D(s)$. Moreover, $\|P_D(y) - P_D(y')\| \leq \|y - y'\|$ for all $y' \in \mathbb{R}^n$.*

(ii) *$P_D(y) \in \operatorname{ri} F$ if and only if $y \in C_F := \operatorname{ri} F + N_D(F) = \{s + w \mid s \in \operatorname{ri} F,\ w \in N_D(F)\}$. In any decomposition $y = s + w$ with $s \in \operatorname{ri} F$ and $w \in N_D(F)$, necessarily $s = P_D(y)$.*

(iii) *If $y \in C_F$, then $P_D(y) = P_{\operatorname{aff} F}(y) = c_F + P_{T_F}\, y$, where $c_F = P_{\operatorname{aff} F}(0) \in T_F^\perp$.*

(iv) *C_F is convex.*

Proof. (i) Let $f(v) = \frac{1}{2}\|v - y\|^2$, which is strictly convex, so that $P_D(y)$ is its unique minimizer on D. For $s, v \in D$, $f(v) = f(s) + \langle s - y, v - s \rangle + \frac{1}{2}\|v - s\|^2$. If $\langle y - s, v - s \rangle \leq 0$ for all $v \in D$, i.e. $y - s \in N_D(s)$, then $f(v) \geq f(s)$ on D. Conversely, if s minimizes f on D, then $0 \leq f(s + \tau(v - s)) - f(s) = \tau \langle s - y, v - s \rangle + \frac{\tau^2}{2}\|v - s\|^2$ for $v \in D$ and $\tau \in (0, 1]$, by convexity of D. Dividing by τ and letting $\tau \to 0$ gives $\langle y - s, v - s \rangle \leq 0$. For the Lipschitz bound, write $s = P_D(y)$ and $s' = P_D(y')$. Adding the inequalities $\langle y - s, s' - s \rangle \leq 0$ and $\langle y' - s', s - s' \rangle \leq 0$ gives $\|s - s'\|^2 \leq \langle y - y', s - s' \rangle \leq \|y - y'\|\, \|s - s'\|$.

(ii) If $s = P_D(y) \in \operatorname{ri} F$, then $w = y - s \in N_D(s) = N_D(F)$ by (i) and Lemma B(iii), and $y = s + w \in C_F$. Conversely, let $y = s + w$ with $s \in \operatorname{ri} F$ and $w \in N_D(F)$. Then $s \in D$ and $y - s = w \in N_D(s)$, so $s = P_D(y)$ by (i), and $P_D(y) \in \operatorname{ri} F$. This also proves the uniqueness of the decomposition.

(iii) Let $E = \operatorname{aff} F = c + T_F$. For $s \in E$ with $y - s \in T_F^\perp$ and any $e \in E$, we have $s - e \in T_F$, hence $\|y - e\|^2 = \|y - s\|^2 + \|s - e\|^2$. So $P_E(y)$ is the unique $s \in E$ with $y - s \in T_F^\perp$. If $y \in C_F$, then $s = P_D(y)$ belongs to $\operatorname{ri} F \subseteq E$ and $y - s \in N_D(F) \subseteq T_F^\perp$ by (ii) and Lemma B(iii), so $P_D(y) = P_E(y)$. For $y = 0$, the characterization gives $c_F = P_E(0) \in T_F^\perp$. Then, for any y, $c_F + P_{T_F} y \in E$ and $y - c_F - P_{T_F} y = (y - P_{T_F} y) - c_F \in T_F^\perp$, so $P_E(y) = c_F + P_{T_F} y$.

(iv) $\operatorname{ri} F$ is convex by Lemma A(ii), and $N_D(F)$ is a convex cone. The sum of two convex sets is convex: $\lambda(s_1 + w_1) + (1 - \lambda)(s_2 + w_2) = (\lambda s_1 + (1 - \lambda) s_2) + (\lambda w_1 + (1 - \lambda) w_2)$ for $\lambda \in [0, 1]$. □

With $y = -tg$, Lemma C(i) and Lemma B(ii) characterize the path: $s(t) \in D$ and

$$tg + s(t) = A^\top \xi(t) + \sum_i \eta_i(t)\, e_i, \qquad \begin{cases} \eta_i(t) \geq 0 & \text{if } s_i(t) = \ell_i^\Delta < u_i^\Delta, \\ \eta_i(t) \leq 0 & \text{if } s_i(t) = u_i^\Delta > \ell_i^\Delta, \\ \eta_i(t) = 0 & \text{if } \ell_i^\Delta < s_i(t) < u_i^\Delta, \end{cases} \tag{1.1}$$

with any sign if $\ell_i^\Delta = u_i^\Delta$. These are the KKT conditions of $\min_{s \in D} \frac{1}{2}\|s + tg\|^2$. With $\sigma_i = +1$ at a lower and $\sigma_i = -1$ at an upper bound, the sign condition reads $\sigma_i \eta_i \geq 0$, as in §2.3 of the plan.

1.3 Pieces, events and multipliers

Proposition 1 (structure of the path)**.**

(i) $[0, \infty)$ *is partitioned into the sets* $I_F = \{t \geq 0 \mid s(t) \in \operatorname{ri} F\}$, F *ranging over the faces of* D*, and each* I_F *is an interval.*

(ii) On I_F, $s(t) = P_{\operatorname{aff} F}(-tg) = c_F - t\, z_F$ *with* $z_F = P_{T_F}\, g$*, so* $s'(t) = -z_F$ *in the interior of* I_F*. The path is continuous and piecewise affine with finitely many pieces* $0 = t_0 < t_1 < \cdots < t_N = T_{\text{end}}$*, and it is constant on* $[T_{\text{end}}, \infty)$*.*

(iii) A piece on $\operatorname{ri} F$ *ends at a* hit*, when a bound* $\beta \notin \mathcal{W}(F)$ *becomes active (a free variable reaches a bound), or at a* release*, when no multiplier vector of the right signs represents* $tg + s(t)$ *any more. When the rows of* A *and the* $e_i^\top$*,* $i \in I(F)$*, are linearly independent, a release means that a multiplier* $\eta_j(t) = \eta_j(t_k) + (t - t_k)\theta_j$*, affine with slope* θ_j *(the bound part of the decomposition* $g = z_F + A^\top y + E_{I(F)}\theta$*), crosses zero with* $\sigma_j \theta_j < 0$*.*

Proof. (i) *Partition.* For $t \geq 0$, $s(t) \in D$ belongs to $\operatorname{ri} F$ for exactly one face F by Lemma A(iii), that is, t belongs to exactly one set I_F. *Intervals.* By Lemma C(ii) with $y = -tg$, $I_F = \{t \geq 0 \mid -tg \in C_F\}$. Let $t_1, t_2 \in I_F$ and $t = \lambda t_1 + (1 - \lambda) t_2$ with $\lambda \in [0, 1]$. Since $t \mapsto -tg$ is linear, $-tg = \lambda(-t_1 g) + (1 - \lambda)(-t_2 g)$, which belongs to C_F by convexity (Lemma C(iv)). Hence $t \in I_F$. A convex subset of $\mathbb{R}$ is an interval, which may be empty or reduced to a point.

(ii) *Affine on each* I_F. Lemma C(iii) with $y = -tg$ gives $s(t) = c_F - tP_{T_F} g$ for $t \in I_F$. *Finitely many pieces.* By Lemma A(iv), the nonempty sets I_F are finitely many pairwise disjoint intervals covering $[0, \infty)$. Let $0 = t_0 < t_1 < \cdots < t_N$ be their finite endpoints. An open interval (t_k, t_{k+1}), or (t_N, ∞), contains no endpoint, so it meets a single I_F: two disjoint intervals meeting it would have an endpoint inside it. Hence s is affine on each of these open intervals. *Continuity.* By Lemma C(i), $\|s(t) - s(t')\| \leq |t - t'|\, \|g\|$, so s is also affine on their closures. *End of the path.* On $[t_N, \infty)$, s is affine and bounded, since D is bounded, so its slope vanishes. We set $T_{\text{end}} = t_N$.

(iii) Let F be a face such that I_F has a nonempty interior and a finite right endpoint $\bar{t}$. For all $t \in \mathbb{R}$, set $\hat{s}(t) = c_F - tz_F$ and $\hat{w}(t) = -tg - \hat{s}(t)$. We claim that, for $t \geq 0$,

$$t \in I_F \iff \hat{s}(t) \in \operatorname{ri} F \quad \text{and} \quad \hat{w}(t) \in N_D(F).$$

If the right-hand side holds, then $-tg = \hat{s}(t) + \hat{w}(t) \in C_F$, and $t \in I_F$ by Lemma C(ii). Conversely, if $t \in I_F$, then $s(t) = \hat{s}(t) \in \operatorname{ri} F$ by (ii), and $\hat{w}(t) = -tg - s(t) \in N_D(F)$ by the uniqueness in Lemma C(ii).

Since $c_F \in \operatorname{aff} F$ and $z_F \in T_F$, $\hat{s}(t) \in \operatorname{aff} F$ for all t. So, by Lemma A(ii), the first condition reads $\langle a_\beta, \hat{s}(t) \rangle < b_\beta$ for all $\beta \notin \mathcal{W}(F)$: finitely many strict affine inequalities in t. The second condition defines a closed set of values of t, since $N_D(F)$ is closed and $\hat{w}$ is continuous. For $t > \bar{t}$ arbitrarily close to $\bar{t}$, $t \notin I_F$, so one of the two conditions fails.

- *Hit.* If the first condition fails, there is a bound $\beta \notin \mathcal{W}(F)$, one of finitely many, with $\langle a_\beta, \hat{s}(t)\rangle \geq b_\beta$ for $t > \bar{t}$ arbitrarily close to $\bar{t}$. Since $\langle a_\beta, \hat{s}(t)\rangle < b_\beta$ in the interior of I_F, continuity gives $\langle a_\beta, s(\bar{t})\rangle = b_\beta$: the bound β, inactive along the piece, becomes active at $\bar{t}$.

- *Release.* If the second condition fails, $\hat{w}(t) \notin N_D(F)$ for $t > \bar{t}$ arbitrarily close to $\bar{t}$, while $\hat{w}(\bar{t}) \in N_D(F)$ since $N_D(F)$ is closed. Now $-\hat{w}(t) = tg + \hat{s}(t) = t(g - z_F) + c_F$, and both $g - z_F = (I - P_{T_F})g$ and c_F belong to $T_F^\perp = \text{range}[A^\top \; E_{I(F)}]$. So $-\hat{w}(t)$ can always be written $A^\top\xi + E_{I(F)}\eta$, but, by Lemma B(ii), for $t > \bar{t}$ close to $\bar{t}$, none of these representations has $\sigma_i\eta_i \geq 0$ for all $i \in I(F)$.

Both events may occur at the same $\bar{t}$, which is then a degenerate breakpoint.

Finally, suppose that the rows of A and the $e_i^\top$, $i \in I(F)$, are linearly independent. Then the coefficients $(\xi(t), \eta(t))$ of $-\hat{w}(t)$ are unique, and they are affine in t since $-\hat{w}$ is affine. With the unique decomposition $g - z_F = A^\top y + E_{I(F)}\theta$, they read $\eta(t) = \eta(\bar{t}) + (t - \bar{t})\,\theta$. If $\sigma_i\eta_i(\bar{t}) > 0$, this sign persists for t close to $\bar{t}$. If $\sigma_i\eta_i(\bar{t}) = 0$ and $\sigma_i\theta_i \geq 0$, then $\sigma_i\eta_i(t) \geq 0$ for $t > \bar{t}$. Hence $\hat{w}(t)$ leaves $N_D(F)$ right after $\bar{t}$ if and only if $\sigma_j\eta_j(\bar{t}) = 0$ and $\sigma_j\theta_j < 0$ for some $j \in I(F)$: the multiplier of the bound of j crosses zero, which is the release of that bound. □

At a breakpoint, the right derivative is $s'(t^+) = P_K(-g)$ with $K = T_D(s) \cap \{tg + s\}^\perp$ [3]. This is what `local_cone_projection!` computes. The proofs below only use Proposition 1.

1.4 Speed along the path

On a piece, $\|s'(t)\| = \|z_F\|$ is the norm of the reduced gradient.

Lemma 2 (hits slow down, releases speed up)**.** *Let F, F' be the faces of two consecutive pieces, and write $z = z_F$, $z' = z_{F'}$.*

- *If $F' \subseteq F$ (a hit), then $T_{F'} \subseteq T_F$ and $\|z'\| \leq \|z\|$.*

- *If $I(F') = I(F) \setminus \{j\}$ (a single release, independent rows), then*

$$\|z'\|^2 = \|z\|^2 + \theta_j^2 \, \|P_{T_{F'}} e_j\|^2 \;\geq\; \|z\|^2.$$

Proof. If $F' \subseteq F$, then $\mathcal{W}(F') \supseteq \mathcal{W}(F)$, so $T_{F'} \subseteq T_F$ by Lemma A(i), and the first claim holds because $z' = P_{T_{F'}} z$. For the second, decompose $g = z + A^\top y + E_{I(F)}\theta$. The vectors $A^\top y$ and e_k, $k \neq j$, are orthogonal to $T_{F'}$, and $P_{T_{F'}} z = z$, so $z' = z + \theta_j P_{T_{F'}} e_j$. Moreover $P_{T_{F'}} e_j \perp T_F$, since its component in T_F is $P_{T_F} e_j = 0$, so the two terms are orthogonal. □

On a box, a coordinate that reaches a bound stays there, so every breakpoint is a hit and $\|s'\|$ is nonincreasing. Over a polyhedron, the dual events make $\|s'\|$ jump up. A stationary piece followed by a release (as in the manuscript's example: $0 \to \sqrt{2}/10$) is the extreme case.

2 Path inequalities

Lemma 3. *For all $t > 0$:*

(a) $L(t) = \int_0^t \|s'(\tau)\|^2 \, d\tau$. *Hence L is continuous, nondecreasing, and constant on $[T_{\text{end}}, \infty)$, where it equals $\max_t L$.*

(b) $\|s(t)\|^2 \leq t\, L(t)$.

(c) $L(t) \geq -\langle g, v\rangle - \dfrac{\|v\|^2}{4t}$ *for every $v \in D$.*

(d) $L(t) \geq \dfrac{h^2}{2} \min\left(1, \dfrac{\Delta}{h}, 2t\right) = \dfrac{h}{2} \min(h, \Delta, 2th)$, *with* $h = h(x)$.

Proof. (a) On a piece, $-\langle g, s' \rangle = \langle g, P_{T_F} g \rangle = \|P_{T_F} g\|^2 = \|s'\|^2$, and $L(0) = 0$.

(b) Take the inner product of (1.1) with $s(t)$. Since $As = 0$, $\langle tg + s, s \rangle = \sum_i \eta_i s_i$. Each term is nonpositive: at a lower bound $s_i = \ell_i^\Delta \leq 0$ and $\eta_i \geq 0$, at an upper bound $s_i = u_i^\Delta \geq 0$ and $\eta_i \leq 0$, and $s_i = 0$ for a fixed variable. Hence $t\langle g, s \rangle \leq -\|s\|^2$. This is also the variational inequality of the projection with $v = 0$.

(c) The projection gives $\langle -tg - s, v - s \rangle \leq 0$, i.e. $t\langle g, s \rangle \leq t\langle g, v \rangle + \langle s, v \rangle - \|s\|^2 \leq t\langle g, v \rangle + \|v\|^2/4$.

(d) Let $c = \min(1, \Delta/\|p\|_\infty) \geq \min(1, \Delta/h)$ and $v = \lambda c p$ with $\lambda \in [0, 1]$. Then $v \in D$, because $\Omega - x$ is convex and contains 0, and $\|v\|_\infty \leq \Delta$. The projection p satisfies $\langle g, p \rangle \leq -\|p\|^2 = -h^2$ (variational inequality with $y = x$). Then (c) gives $L(t) \geq \lambda c h^2 - \lambda^2 c^2 h^2/(4t)$. Take $\lambda = 1$ if $2t \geq c$ and $\lambda = 2t/c$ otherwise. □

Item (d) deals with open point 2 of the plan: merging the ℓ_∞ trust region into the bounds costs nothing, since a scaled projected gradient step stays in D.

Lemma 3(a) shows why the decrease is governed by the *integral* of $\|s'\|^2$. The pointwise value of the reduced gradient at early times does not control it, unless $\|s'\|$ is monotone.

3 The first local minimizer

Scan the pieces in order. On a piece $[t_i, t_{i+1}]$ with $d = s' \neq 0$, let $\varphi_1 = \langle g + Hs(t_i), d \rangle = \psi'(t_i^+)$ and $\varphi_2 = \langle d, Hd \rangle$. Stop at t_i if $\varphi_1 \geq 0$, and at $t_i - \varphi_1/\varphi_2$ if $\varphi_1 < 0 < \varphi_2$ and this point lies before t_{i+1}. Otherwise, go to the next piece. Stationary pieces ($d = 0$) are skipped, and the scan stops at T_{end} at the latest. Call t_1 the stopping point. By construction, ψ is nonincreasing on $[0, t_1]$. If $t_1 < T_{\text{end}}$, the direction $d \neq 0$ at t_1 (of the current piece, or of the next nonstationary one) satisfies $\langle g + Hs(t_1), d \rangle \geq 0$. Hence $\|d\|^2 = -\langle g, d \rangle \leq \langle Hs(t_1), d \rangle$, that is

$$\|d\| \leq \|H\| \, \|s(t_1)\|. \tag{3.1}$$

So an early stop requires a slow direction.

3.1 Without releases: the box argument is correct

Proposition 4. *Suppose $\|s'\|$ is nonincreasing on (t_1, ∞), i.e. no bound is released after t_1. This holds in particular when Ω is a box, with the trust region included in the bounds. Then*

$$-\psi(t_1) \;\geq\; \frac{3}{10}\, h \, \min\Big(\Delta,\; \frac{h}{1 + \|H\|}\Big).$$

Proof. Let $\tau_0 = 1/(2(1 + \|H\|))$, so that $\|H\|\tau_0 \leq \frac{1}{2}$. By Lemma 3(b), for $t \leq \tau_0$, $-\psi(t) \geq L(t)(1 - \frac{1}{2}\|H\|t) \geq \frac{3}{4} L(t)$.

- If $t_1 \geq \tau_0$: ψ is nonincreasing on $[0, t_1]$, so $-\psi(t_1) \geq -\psi(\tau_0) \geq \frac{3}{4} L(\tau_0)$.
- If $t_1 = T_{\text{end}} < \tau_0$: $-\psi(t_1) \geq \frac{3}{4} L(T_{\text{end}}) \geq \frac{3}{4} L(\tau_0)$.
- Otherwise $t_1 < \min(\tau_0, T_{\text{end}})$. By monotonicity of $\|s'\|$ and (3.1),

$$L(\tau_0) - L(t_1) = \int_{t_1}^{\tau_0} \|s'\|^2 \leq \tau_0 \|d\|^2 \leq \tau_0 \|H\|^2 \|s(t_1)\|^2 \leq \tau_0 \|H\|^2 t_1 L(t_1) \leq \tfrac{1}{4} L(t_1),$$

  so $-\psi(t_1) \geq \frac{3}{4} L(t_1) \geq \frac{3}{5} L(\tau_0)$.

In all cases $-\psi(t_1) \geq \frac{3}{5} L(\tau_0)$. Conclude with Lemma 3(d) and $2\tau_0 h = h/(1 + \|H\|)$. □

The monotonicity of the reduced gradient, the core of [1] and of the old draft, is used in exactly one place: to stop the path from speeding up after t_1. This is where releases break the argument.

3.2 With releases: a counterexample

Take the manuscript's polyhedron plus a free variable:

$$\Omega = \{x \in \mathbb{R}^4 \mid x_1 + x_2 + x_3 = 0,\ x_1 \geq -1,\ x_2 \leq 1\},$$
$$g = (0.3,\ -0.8,\ 0.5,\ a), \qquad H = vv^\top, \quad v = e_4 - e_2,$$

with $\Delta = 1$, $x_\delta = (-1, 1, 0, 0) + \delta(1, -3, 2, 0)$ and $0 < a \leq \delta \leq 0.01$. Then $H \succeq 0$ and $\|H\| = 2$. Since $D = D_3 \times [-1, 1]$, the path is the product of the 3D path of the manuscript's zigzag example and $s_4(t) = -at$ (Table 1).

Table 1: Pieces of the path of the counterexample.

t	direction s'	event at the end
$[0,\ 10\delta/3]$	$-g$	hit $x_1 \geq -1$
$[10\delta/3,\ 50\delta/13]$	$(0, 0.65, -0.65, -a)$	hit $x_2 \leq 1$: vertex
$[50\delta/13,\ 5\delta]$	$(0, 0, 0, -a)$, **slow**	release of $x_1 \geq -1$
$[5\delta,\ \approx 10]$	$(0.1, 0, -0.1, -a)$	trust region

At the vertex, $g = (0, 0, 0, a) + 0.5\,(1, 1, 1, 0) - 0.2\,e_1 - 1.3\,e_2$. The multiplier of $x_1 \geq -1$ has slope $\theta_1 = -0.2 < 0$: it decreases from $0.45 \cdot 20\delta/39$ and vanishes at $t = 5\delta$. The speed jumps from a to $\sqrt{0.02 + a^2}$ (Lemma 2).

- **Before the vertex** the scan does not stop. On the first two pieces $\varphi_1 \leq -0.845 + 1.8\delta < 0$, and the minimizers of the quadratics lie beyond the pieces.
- **At the vertex** $t_v = 50\delta/13$, $s = (-\delta, 3\delta, -2\delta, -at_v)$, and on the slow piece

  $$\psi'(t_v^+) = -a^2 + \langle v, s\rangle\langle v, d\rangle = a\,(3\delta + at_v - a) > 0.$$

  So $t_1 = t_v$, and the decrease is

  $$-\psi(t_1) = L(t_v) - \tfrac{1}{2}\langle v, s\rangle^2 = 3.7\,\delta + \tfrac{50}{13}a^2\delta - \tfrac{1}{2}\big(3\delta + \tfrac{50}{13}a\delta\big)^2 = 3.7\,\delta + O(\delta^2).$$

- **The measures stay bounded away from zero:** $h(x_\delta) \to \sqrt{0.02 + a^2} \geq \sqrt{2}/10$, $\chi(x_\delta) \to 0.2$, and $\pi_\varepsilon(x_\delta) = \max(a, 0.2) = 0.2$ as soon as both bounds are ε-active.
- **After the release** ψ keeps decreasing by about 0.2 up to the trust region ($\psi' = -(0.02 + a^2) + a(3\delta + at) < 0$).

Proposition 5. *No inequality $-q(s^C) \geq \Phi(h(x), \Delta, \|H\|)$, with $\Phi > 0$ for positive arguments, can hold for the first local minimizer over polyhedra. The same holds with π_ε or χ in place of h, and even for convex models.*

Table 2 gives the values computed with `cauchy_step!` (current code) for $a = \delta/2$.

With $a = \delta^2$, the slow piece falls below `tol_dir` once $\delta \leq 10^{-4}$. The code then treats it as stationary and skips it, which shows that the slow piece is a perturbation of the stationary piece of the zigzag example. Skipping stationary pieces handles $a = 0$, and the safeguard below handles $a > 0$.

This is a failure of the per-iteration decrease, which the trust-region analysis needs. It is not a proven failure of the algorithm: here the Cauchy point lands on the vertex, and from there the next path starts with the release direction. A zigzag sequence would have to repeat this near ever closer faces. Whether one exists is open, and the safeguard makes the question moot.

Table 2: Counterexample with $a = \delta/2$. The last column is the best value of ψ on $[0, \hat{t}]$ for $\kappa_G = \frac{1}{2}$ (Section 4), evaluated on the traced path.

δ	$-q(s^C)$, code	t_1	$h(x_\delta)$	$\chi(x_\delta)$	$-q$, safeguarded
1e-1	3.20e-1	3.85e-1	0.397	0.580	0.326
1e-2	3.66e-2	3.85e-2	0.146	0.238	0.230
1e-3	3.70e-3	3.85e-3	0.1415	0.204	0.203
1e-4	3.70e-4	3.85e-4	0.1414	0.200	0.200
1e-6	3.70e-6	3.85e-6	0.1414	0.200	0.200

4 A safeguarded Cauchy point

4.1 Definition

Fix $\kappa_G \in (0, 1)$. Let

$$t_G = \inf\{t > 0 \mid \psi(t) \geq -\kappa_G\, L(t)\} \in (0, \infty], \qquad \hat{t} = \min(t_G, T_{\text{end}}).$$

t_G is where the model decrease falls to the fraction κ_G of the linear decrease: the curvature starts to matter. Before t_G, $\psi \approx -L$, and a stop there can only come from a slow piece, by (3.1).

Definition (safeguarded Cauchy point)**.** The Cauchy step $s^C = s(t^C)$ satisfies **condition (C)** if $\psi(t^C) \leq \psi(\hat{t})$. The following **rule** produces such a point.

- If $t_1 \geq \hat{t}$, take $t^C = t_1$. This is the common case, and the current behavior.
- Otherwise (t_1 is *premature*), go on scanning from $\hat{t}$ to the next stop $t^\sharp$: the first $t \geq \hat{t}$ with $\psi'(t^+) \geq 0$ on the next nonstationary piece, or T_{end}. Take for t^C the best of the stops met on $[0, t^\sharp]$.

The rule satisfies (C). In the first case, ψ is nonincreasing on $[0, t_1] \ni \hat{t}$. In the second, ψ decreases on $[\hat{t}, t^\sharp]$, so $\psi(t^C) \leq \psi(t^\sharp) \leq \psi(\hat{t})$. Any other way of reaching (C) is also fine for the theory.

The test $t_1 \geq \hat{t}$ means that $t_1 = T_{\text{end}}$ or that $r(t) = \psi(t) + \kappa_G L(t) \geq 0$ for some $t \in (0, t_1]$. On a piece,

$$r(t_i + \tau) = r(t_i) + (\varphi_1 - \kappa_G \langle g, d\rangle)\, \tau + \tfrac{1}{2}\varphi_2 \tau^2$$

(note $L' = -\langle g, d\rangle$), so the scan only needs the running value of L, one inner product it already computes, and the maximum of a quadratic on an interval. This is in the spirit of the generalized Cauchy point of [2, §12.2], with a single Goldstein condition: the trust region is part of D, so their condition on $\|s\|$ versus Δ is replaced by the end of the path.

If $\kappa_G \geq \frac{1}{2}$, a minimizer inside the first piece is always accepted: there $\psi(t) = -t\|d\|^2 + \frac{1}{2}t^2\langle d, Hd\rangle$ and $L(t) = t\|d\|^2$, so $\psi(t_1) = -\frac{1}{2}L(t_1) \geq -\kappa_G L(t_1)$.

4.2 Decrease

Lemma 6. *If x is not critical, then $t_G \geq 2(1 - \kappa_G)/\|H\|$ (with $t_G = \infty$ if $H = 0$) and $\psi(\hat{t}) \leq -\kappa_G L(\hat{t})$.*

Proof. If x is not critical, $s(t) \neq 0$, hence $L(t) \geq \|s(t)\|^2/t > 0$, for all $t > 0$. By Lemma 3(b),

$$\psi(t) + \kappa_G L(t) \leq -(1 - \kappa_G)L(t) + \tfrac{1}{2}\|H\| t L(t) < 0 \qquad \text{for } t < 2(1 - \kappa_G)/\|H\|.$$

For $t < t_G$, $\psi(t) < -\kappa_G L(t)$. If $\hat{t} = T_{\text{end}} < t_G$, the claim is this inequality at T_{end}. If $\hat{t} = t_G < \infty$, continuity gives $\psi(t_G) = -\kappa_G L(t_G)$. □

Theorem 7 (Cauchy decrease)**.** *If $s^C = s(t^C)$ satisfies (C), then*

$$-q(s^C) \;\geq\; \frac{\kappa_G}{2}\, h(x)\, \min\Big(\Delta,\; \frac{\kappa'\, h(x)}{1+\|H\|}\Big), \qquad \kappa' = \min(1, 4(1-\kappa_G)).$$

Proof. Assume $h = h(x) > 0$ and let $\tilde{t} = 2(1-\kappa_G)/(1+\|H\|) \leq t_G$. Since L is nondecreasing and maximal from T_{end} on, $L(\hat{t}) \geq L(\tilde{t})$ in both cases $\hat{t} = t_G$ and $\hat{t} = T_{\text{end}}$. By (C), Lemma 6 and Lemma 3(d),

$$-\psi(t^C) \geq -\psi(\hat{t}) \geq \kappa_G L(\hat{t}) \geq \kappa_G L(\tilde{t}) \geq \frac{\kappa_G h}{2} \min(h, \Delta, 2\tilde{t}h),$$

and $\min(h, 2\tilde{t}h) \geq \kappa' h/(1+\|H\|)$. □

The theorem uses neither the rank of the active rows nor ε, and has no dimension constant. Only the convexity of D, $0 \in D$, and the piecewise structure are used.

5 Convergence of the inner iterations

The steps satisfy $s_k \in D_k$ and $q_k(s_k) \leq q_k(s_k^C)$ (minor iterations), and H_k changes only after successful iterations. The assumptions are those of the manuscript:

- r and c are twice continuously differentiable (`assumption:functions_C2`);
- the level set $\mathcal{X} = \{x \in \Omega \mid \varphi(x) \leq \varphi(x_0)\}$ is compact (`assumption:inner_iterates_compact`);
- $\lim_k b_k(\varphi_{k+1} - \varphi_k) = 0$, with $b_k = 1 + \max_{i \leq k} \|H_i\|$ (`assumption:hessian_norm_compared_convergence_speed`).

Theorem 8. *If Algorithm 2, with the Cauchy step of Section 4, generates infinitely many iterates, then* $\liminf_k h(x_k) = 0$.

Proof. If $h(x_k) \geq \underline{h} > 0$ for $k \geq k_0$, Theorem 7 and $b_k \geq \max(1, 1+\|H_k\|)$ give

$$-q_k(s_k) \geq c \min(\Delta_k, \zeta/b_k), \qquad c = \kappa_G \underline{h}/2, \quad \zeta = \kappa' \underline{h}.$$

This is inequality `eq:decrease_bk`. The proof of `theo:inner_convergence` uses nothing else about the Cauchy step, and applies word for word. □

Theorem 9. *Under the same assumptions,* $\lim_k h(x_k) = 0$.

Proof. We follow [1, Theorem 11].

1. *Setup.* Suppose $h(x_{m_i}) \geq \epsilon_1 > 0$ along a subsequence, and let $l_i > m_i$ be the first index with $h(x_{l_i}) < \epsilon_2 = \epsilon_1/2$ (Theorem 8). For the successful $k \in [m_i, l_i)$, Theorem 7 gives $\varphi_k - \varphi_{k+1} \geq \eta_1 \frac{\kappa_G \epsilon_2}{2} \min(\Delta_k, \kappa' \epsilon_2 / b_k)$.
2. *Steps are short.* Since $b_k(\varphi_k - \varphi_{k+1}) \to 0$, we get $b_k \Delta_k \to 0$ on these iterations, so the minimum is Δ_k for i large. Hence $\|x_{k+1} - x_k\| \leq \sqrt{n}\, \Delta_k \leq C(\varphi_k - \varphi_{k+1})$.
3. *Contradiction.* Summing, $\|x_{m_i} - x_{l_i}\| \leq C(\varphi_{m_i} - \varphi_{l_i}) \to 0$, since (φ_k) is nonincreasing and bounded below on $\mathcal{X}$. The map $x \mapsto \|P_\Omega(x - \nabla\varphi(x)) - x\|$ is uniformly continuous on the compact set $\mathcal{X}$, so $h(x_{m_i}) - h(x_{l_i}) \to 0$, which contradicts $h(x_{m_i}) - h(x_{l_i}) \geq \epsilon_1/2$. □

The manuscript now only claims $\liminf \pi_\varepsilon = 0$, because π_ε is not continuous. Theorem 9 gives a full limit for h.

6 Link with the stopping test

Recall that $\mathcal{N}_\varepsilon(x)$ is generated by $\pm$ the rows of A, by $-e_i$ ($i \in \mathcal{A}_\varepsilon^-(x)$) and by e_i ($i \in \mathcal{A}_\varepsilon^+(x)$), and that $\pi_\varepsilon(x) = \max(\|u\|_\infty, \|\theta^{\mathrm{w}}\|_\infty)$, with $u = P_{T_\varepsilon(x)}g$ and θ a bound part of a decomposition $g = u + A^\top\xi + E_S\theta$, $S = \mathcal{A}_\varepsilon(x)$. When the rows of $\tilde{A}_S = [A; E_S^\top]$ are dependent, θ is chosen to minimize $\|\theta^{\mathrm{w}}\|_2$ over all decompositions, which is what `wrong_sign_redundant` does: its parametrization by the redundant bounds spans the whole null space of $[A^\top\ E_S]$.

Lemma 10 (link)**.** *If $\|p\|_\infty \le \varepsilon$, then* $\operatorname{dist}(-g, \mathcal{N}_\varepsilon(x)) \le \|p\| = h(x)$.

Proof. Let $y = x + p$. If $y_i = l_i$, then $x_i - l_i = -p_i \le \varepsilon$, so $i \in \mathcal{A}_\varepsilon^-(x)$, and likewise for the upper bounds. Hence $\mathcal{N}_\Omega(y) \subseteq \mathcal{N}_\varepsilon(x)$. Moreover $-g - p = (x - g) - y \in \mathcal{N}_\Omega(y)$. □

Lemma 11. *Let $\sigma^+ > 0$ be the smallest nonzero singular value of the finitely many matrices $\tilde{A}_S$, $S \subseteq \{1, \dots, n\}$. With the multipliers of minimal violation, for all $x \in \Omega$, $\varepsilon \ge 0$ and g,*

$$\pi_\varepsilon(x) \le \max(1, 1/\sigma^+)\ \operatorname{dist}(-g, \mathcal{N}_\varepsilon(x)).$$

Proof.

1. *Moreau decomposition.* Write $-g = d + w$ with $d = P_{\mathcal{T}_\varepsilon}(-g)$, $w \in \mathcal{N}_\varepsilon$ and $\|d\| = \operatorname{dist}(-g, \mathcal{N}_\varepsilon(x))$. Then $w = A^\top\xi' - E_S\theta^+$, where θ^+ has the right signs (the sign is free for fixed variables).

2. *Reduced gradient.* $u = P_{T_\varepsilon}g = -P_{T_\varepsilon}d$, so $\|u\| \le \|d\|$.

3. *Multipliers.* $d - P_{T_\varepsilon}d = \tilde{A}_S^\top y$ with y of minimal norm, hence $y \in \operatorname{range}\tilde{A}_S$ and $\|y\| \le \|d\|/\sigma^+$. Then $g = u + A^\top(-\xi' - y_A) + E_S(\theta^+ - y_S)$ is a valid decomposition, and its violation is at most $|y_{S,i}|$ componentwise. So the minimal violation satisfies $\|\theta^{\mathrm{w}}\|_\infty \le \|\theta^{\mathrm{w}}\|_2 \le \|y_S\| \le \|d\|/\sigma^+$. □

The right inequality of `lemma:certification`, $\operatorname{dist}(-g, \mathcal{N}_\varepsilon(x)) \le \sqrt{n}\,\pi_\varepsilon(x)$, holds for any multipliers, so it is unchanged. The left one (with $\sigma_{\min}$ and Assumption `full_rank_active`) is replaced by Lemma 11.

Corollary 12. *For every $\varepsilon > 0$ and $\omega > 0$, the stopping test $\pi_\varepsilon(x_k) \le \omega$ of Algorithm 2 is met after finitely many iterations.*

Proof. By Theorem 8 there is k with $h(x_k) \le \min(\varepsilon, \omega/\max(1, 1/\sigma^+))$. Then $\|p_k\|_\infty \le h(x_k) \le \varepsilon$, and Lemmas 10 and 11 give $\pi_\varepsilon(x_k) \le \max(1, 1/\sigma^+)\,h(x_k) \le \omega$. □

The only requirement on ε is $\varepsilon > 0$. The constant does not depend on ε, so $\varepsilon_K = \min(\varepsilon_{\max}, \omega_K)$ works for any $\varepsilon_{\max} > 0$. The outer analysis (§3.2 of the manuscript) is unchanged.

7 Consequences

7.1 Manuscript (`sn-preprint-theorie.tex`)

- **§2.1.** Remove Assumption `full_rank_active` and the threshold $\bar{\varepsilon}$, and define the multipliers of minimal violation.
- **§2.3.** The Cauchy step is the first local minimizer along the exact path `eq:projected_gradient_path`, with the safeguard of Section 4.1 (one more constant, κ_G).
- **§3.1.** Replace `lemma:eps_threshold`, `prop:computed_path`, `lemma:initial_direction` and `theo:cauchy_decrease` by Proposition 1, Lemmas 2 and 3, Lemma 6 and Theorem 7.

  - Proposition 4 (box) and the example of Section 3.2 can replace the zigzag remark. They explain why stationary pieces are skipped, and why the first local minimizer needs the safeguard over a polyhedron and not over a box.
  - `theo:inner_convergence` keeps its proof, with h in place of $\kappa_0\pi_\varepsilon$. Add Theorem 9, Lemma 10 and Corollary 12.

- **`lemma:certification`.** Replace the left inequality by Lemma 11, and update the paragraph after it ("if $\varepsilon_{\max} \le \bar{\varepsilon}\dots$").
- **§3.2.** Remove "Consistently with the well-posedness of the projector, $\tilde{A}_*$ has full row rank". $T(x_*) = \operatorname{null} \tilde{A}_*$ always has an orthonormal basis.
- **Appendix A.** Add the safeguard to Algorithm `algo:cauchy_point`.
- **Bibliography.** [1] (Theorem 11) and [2] (§12.2) are already there. Add [3] if the critical cone is mentioned.

7.2 Code (`cauchy_step!`, `SubspaceProjector` method)

The box method (`CoordinateSubspaceProjector`) needs no change, by Proposition 4. Algorithm 1 lists the changes to the polyhedral loop, $\bar{\tau}$ being the length of the current piece in t.

Algorithm 1 Safeguarded scan (changes to the polyhedral loop of `cauchy_step!`)

$L \leftarrow 0$, *crossed* $\leftarrow$ false, *best* $\leftarrow$ none ▷ $L = -g^\top s$; *crossed* $\iff$ $t_G \le t$
for each piece **do**
  (unchanged: stationary pieces, parking, next event at distance $\bar{\tau}$)
  if the piece is not stationary **then**
    $w \leftarrow Hd$; $\varphi_1 \leftarrow g^\top d + s^\top w$; $\varphi_2 \leftarrow d^\top w$; $\ell \leftarrow -g^\top d$
    $\tau_{\text{stop}} \leftarrow 0$ if $\varphi_1 \ge 0$;
    $\tau_{\text{stop}} \leftarrow -\varphi_1/\varphi_2$ if $\varphi_2 > 0$ and $-\varphi_1/\varphi_2 < \bar{\tau}$; none otherwise
    crossed $\leftarrow$ *crossed* or
    $\max_{0<\tau\le\min(\tau_{\text{stop}},\bar{\tau})} [pred + \kappa_G L + (\varphi_1 + \kappa_G\ell)\tau + \varphi_2\tau^2/2] \ge 0$
    if $\tau_{\text{stop}} \ne$ none **then**
      record $(t + \tau_{\text{stop}},\ s + \tau_{\text{stop}} d,$ value, working set$)$ in *best* if better
      if *crossed* **then break** ▷ usual case: unchanged result
      ▷ premature: continue; the rest of the piece also updates *crossed*
  if end of the path **then** record the current point in *best*; **break**
  $L \leftarrow L + \ell\bar{\tau}$; $pred \leftarrow pred + \varphi_1\bar{\tau} + \varphi_2\bar{\tau}^2/2$; … ▷ move to the breakpoint, unchanged
return *best*, restoring its working set in `proj_op` if it is not the last point

A small `eps_slope`-style tolerance on the crossing test is enough. The example of Section 3.2 makes a good regression test: the decrease must stay above about 0.2 as $\delta \to 0$.

7.3 Numerical evidence

The test problems are random polyhedra with $(n, m) \in \{(10, 3), (40, 8)\}$, A Gaussian, box $[-1, 1]$ and $x = P_\Omega(z)$ with $z \sim 3\mathcal{N}(0, I)$, so many bounds are active. The other data are $g \sim 3\mathcal{N}(0, I)$, $H = J^\top J$ with J of scale 0.1, 1 or 3, and $\Delta \in \{0.1, 0.5, 2\}$. The code examines a median of 2 pieces (at most 35) up to t_1. Table 3 summarizes 299 paths.

Table 3: Premature first local minimizers and cost of the safeguard over 299 random paths.

κ_G	premature t_1	worst $q(s^C_{\text{code}})/q(s^C_{\text{safe}})$	extra pieces (median / max)
0.60	62	0.993	2 / 14
0.75	53	0.993	1 / 14
0.90	33	0.997	1 / 7
0.99	8	1.000	1 / 5
0.999	0	1.000	—

- **On generic problems the first local minimizer is right.** The safeguard almost never improves the step: it only buys a guarantee against the adversarial geometry of Section 3.2.
- **Choice of κ_G.** A value close to 1, say 0.99, keeps the extra cost negligible. The theoretical constant scales like $\kappa_G(1-\kappa_G)$. Any $\kappa_G \geq \frac{1}{2}$ also accepts a minimizer inside the first piece.
- **Checks on the same instances.** Theorem 7 held with a margin of at least 4.1. The χ version, $\frac{\kappa_G \chi}{2}\min(\Delta, 1, 4(1-\kappa_G)\chi/(n(1+\|H\|)))$, held with a margin of at least 2.4. The inequality $\|s\|^2 \leq tL$ held to rounding, and L was nondecreasing.

8 Remarks

1. **Why not keep the pure first local minimizer?** Its guarantee needs the speed not to increase after t_1, which only the future of the path can tell (Section 3.1). A safeguard has to look ahead somehow. The Goldstein crossing is the cheapest look-ahead with a proof, and it is already computable from the quantities of the scan.
2. **Stationary pieces** are the limit $a = 0$ of Section 3.2. With the safeguard, skipping them is still the right choice: ψ and L are constant there, so they never trigger the crossing.
3. **Number of pieces.** The safeguard can only lengthen the scan up to $t^\sharp$, and still finitely (Proposition 1). The worst case is still the number of faces.
4. **What is not covered.** Rounding tolerances, parking and the `maxpieces` cap. On the cap, the current s still satisfies (C) if the crossing has been seen; otherwise the guarantee is lost, as before. Failure of the local cone projection only affects the direction, not the path theory, since d stays a feasible descent direction.

A Reproducing the counterexample

With the helpers of `test/cauchy_path.jl` (`cauchy_path`, `proj_dykstra`):

```
A = [1.0 1.0 1.0 0.0]
xlow, xupp = [-1.0, -Inf, -Inf, -Inf], [Inf, 1.0, Inf, Inf]
delta = 1e-4; a = delta / 2
x = [-1.0, 1.0, 0.0, 0.0] + delta * [1.0, -3.0, 2.0, 0.0]
g = [0.3, -0.8, 0.5, a]
v = [0.0, -1.0, 0.0, 1.0]
res = cauchy_path(A, xlow, xupp, x, g, 1.0; J = reshape(v, 1, 4))
res.pred                                                  # about -3.7e-4
norm(proj_dykstra(A, xlow .- x, xupp .- x, -g))           # h(x), about 0.1414
```

References

[1] A. R. Conn, N. I. M. Gould, and Ph. L. Toint. Global convergence of a class of trust region algorithms for optimization with simple bounds. *SIAM Journal on Numerical Analysis*, 25(2):433–460, 1988.

[2] A. R. Conn, N. I. M. Gould, and Ph. L. Toint. *Trust-Region Methods*. SIAM, Philadelphia, PA, USA, 2000.

[3] A. Haraux. How to differentiate the projection on a convex set in Hilbert space. Some applications to variational inequalities. *Journal of the Mathematical Society of Japan*, 29(4):615–631, 1977.